

Siva G Nair

AI Engineer | Generative AI • Computer Vision • NLP

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PROFESSIONAL SUMMARY

AI Engineer specializing in Generative AI, Computer Vision, and NLP, with hands-on experience building and deploying industrial and enterprise AI systems. Engineered RAG and LLM-based document automation workflows, PyTorch/OpenCV visual inspection systems, and AWS data pipelines, delivering 95% defect detection accuracy, 90% document classification accuracy, 40% improved information retrieval, and processing 1M+ records. Proficient in Python, SQL, PyTorch, TensorFlow/Keras, OpenCV, RAG, LLMs, embeddings, vector databases, and cloud-based AI deployment.

SELECTED IMPACT

- **95% Defect Detection Accuracy:** Engineered PyTorch/OpenCV CNN-based industrial visual inspection systems.
- **90% Document Classification Accuracy:** Applied LLM-based document processing while reducing client feedback cycles by 30%.
- **1M+ Records Processed:** Optimized AWS data pipelines for real-time model training and inference.

PROFESSIONAL EXPERIENCE

Ombrulla

AI Engineer

Dec 2024 – May 2025

Kochi, Kerala

- Designed and deployed industrial AI proof-of-concepts for computer vision and NLP, incorporating LLM-based document processing and achieving 90% automated document classification accuracy.
- Engineered a RAG pipeline for enterprise document question answering, tabular-data interaction, and automated summarization, improving information retrieval efficiency by 40%.
- Integrated camera systems with PyTorch CNNs and OpenCV for visual inspection, enabling local model training and deployment with 95% defect detection accuracy.
- Streamlined Python, Pandas, and NumPy data pipelines on AWS for real-time processing of 1M+ records, supporting model training and inference.

Luminar Technolab

Data Science Intern

May 2024 – Nov 2024

Kochi, Kerala

- Developed and optimized predictive analytics models with Scikit-learn, TensorFlow, and Keras, improving time-series forecasting accuracy by 25%.
- Preprocessed, cleaned, and feature-engineered datasets exceeding 500K records using Pandas and NumPy to improve model-training inputs.
- Created stakeholder-facing dashboards and visualizations with Power BI, Matplotlib, and Seaborn, reducing decision-making time by 20%.

PROJECTS

RAG-Based Email Automation Agent

GenAI • OCR • Vector Search

- Engineered a multimodal RAG workflow, integrating EasyOCR, PyTorch, Ollama, and Milvus, reducing manual processing time by 40%.
- Implemented event detection and notification triggers, increasing user engagement by 25% through enhanced response efficiency.
- Deployed local LLM using Ollama for private email analysis, achieving 95% accuracy in automated response generation.

Automatic Mail Handling Agent

RAG • GenAI • OCR

- Developed an automated email-handling pipeline that fetches emails, preprocesses content and attachments, and extracts contextual information using RAG and LLMs.
- Implemented OCR for image-based email attachments and incorporated extracted text into the retrieval pipeline.
- Generated context-aware email responses using retrieved information and email content.

Natural Language Database Q&A Agent

RAG • LLM • PostgreSQL

- Engineered a RAG-based framework agent that provides database schema and contextual information to an LLM for accurate data retrieval.
- Integrated PostgreSQL to enable natural-language querying of database information.
- Implemented LLM-based SQL generation, converting plain-English questions into SQL queries and retrieving relevant database records.

Botanic Vision – Rice Plant Disease Detection App

Flask • TensorFlow • Computer Vision

- Developed a Flask/TensorFlow web app identifying five major rice plant diseases with 93% accuracy

Text Emotion Detection

NLP • Keras • Streamlit

- Built a Streamlit NLP application for five-class emotion classification with 85% accuracy using Python, NLTK, and Keras.

CORE COMPETENCIES

- **Languages:** Python, SQL, JavaScript,Java
- **AI/ML & GenAI:** PyTorch, TensorFlow, Keras, OpenCV, RAG, LLMs, Prompt Engineering, Embeddings, AI Agents, Hugging Face
- **LLM & Retrieval:** LangChain, LlamaIndex, Ollama, Milvus, Vector Databases, EasyOCR
- **Data & Databases:** Pandas, NumPy, Scikit-learn, PostgreSQL, MongoDB, Power BI, Matplotlib, Seaborn
- **Deployment & Integration:** AWS, Docker, Flask, Streamlit, Heroku, Cloudflare, Git, GitHub, Model Context Protocol (MCP)

EDUCATION

Sree Buddha College of Engineering,
APJ Abdul Kalam Technological University

2020 – 2024

Bachelor of Technology (B.Tech), Computer Science (Artificial Intelligence and Machine Learning)